

Seasonal Wardrobe Swap SOP (1x Episode: four out, four in, in situ)

One episode swaps four items at the wardrobe bench where they live. The four out-of-season items go first. They are folded, one at a time, and laid in the box. Then the four in-season items come out of the box. Two go on the rail. Two go on the shelf. The episode ends when the box is closed and stowed in its bay. Never split the four out and the four in into two episodes.

The base is passive. It has no motor. You push it by hand to the bench in front of the wardrobe and lock it there. Nothing is carried away to another room. Everything the episode touches is already there when recording starts: the bench, the out stack at the front left, the open box at the front right with the four in-season items in it, the lid past the box, the empty rail above the left end, the empty shelf above the right end, and the empty stow bay under the shelf.

The wardrobe is used as found. The rail is empty. The shelf is empty. The stow bay is empty. The box stands open on the bench, its lid off and lying flat beside it. The bench top is clear but for the out stack, the box, and the lid. The episode ends with four items folded in the box, two items hanging on the rail, two items lying on the shelf, and the box closed and stowed.

This is an in situ task. Four rules come from that. First, everything happens on this bench. Nothing is carried away and nothing new is brought in. Second, every carry goes along the front lane, low and level. You lift a thing just off what it stands on, bring it forward into the lane, and carry it along the lane close over the bench. The only lift that goes up is the hook going onto the rail. Third, the fixtures never move. You never lean on, push, drag, or lift the bench, the rail, the shelf, or the stow bay. Fourth, nothing is ever dragged. Every pick lifts straight up off the surface before it moves anywhere.

Clothes go on the bench, in the box, on the rail, or on the shelf, and nowhere else. Nothing ever touches the floor. An item that lands on the floor is a drop. You do not pick a drop back up. You report it.

The order never changes: fold all four out items into the box; take the two hang items out and hang them; take the two shelf items out and shelve them; put the lid on; stow the box. No in-season item comes out of the box until all four folded items are in it.

Only one thing is held at a time. Nothing is passed from one grip to another, and no grip is changed while an item is off the bench. Every item is put down flat and let go under control.

Setup

Do the base positioning and both checklists before you start an episode.

Cell configuration

Environment camera: 900 mm.

cell_type: bimanual

Base positioning

The base is passive. It has no motor. You push it into place by hand. You never steer it, nudge it, or move it after recording starts. You park it once, before recording, and it stays there until the episode is over.

1. Push the base by hand up to the bench. Stop it square to the front edge of the bench, so the front edge runs straight across the frame and both ends look the same distance away.
2. Stop it in the middle of the bench, so the out stack at the left and the box at the right are the same distance from the middle of the base.
3. Stop it close enough that a gripper reaches the middle of the fold square without stretching the arm. Stop it far enough that no arm, wrist, or part of the base touches the bench front, the rail, or the shelf while both arms work.
4. Check the height. With the base parked, a gripper lifts the top item off the out stack, brings it forward into the front lane, and carries it to the fold square. The wrist and the forearm must not hit the rail or the shelf.
5. Check the rail. A gripper lifts a hanger hook up onto the rail and lets go, without stretching and without hitting the shelf or the other arm.
6. Check the bench. A gripper reaches the out stack, the fold square, both halves of the box, the lid, the shelf, and the stow bay. It does all of them without stretching and without knocking anything over.
7. Lock the base. Push it once, hard, by hand. It must not roll, slide, or turn.
8. If any line from 1 to 6 fails, push the base to a new spot by hand and start again at line 1. Do not work at a bench the arms cannot reach easily.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the bench hard enough to push it. Nobody touches it by hand. You never move it during the task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera points at the middle of the fold square. The frame shows the whole bench top, the out stack, the fold square, the open box with both its halves, the lid, the rail, the shelf, and the stow bay.
3. The camera sees the bench from the front, so you can see where a thing lies and whether it lies flat and square.

4. The camera sees the inside of the box from the front, so you can see how many items are in each half.
5. The camera sees the rail from the front, so you can see whether a hook sits all the way over it.
6. The camera sees the front edge of the shelf, so you can see whether a folded item hangs over it.
7. Both arms are at home with the grippers open.
8. The arms reach the out stack, the fold square, both halves of the box, the lid, the rail, the shelf, and the stow bay. They reach all of them without stretching to a joint limit.
9. Two grippers reach the fold square at the same time, one at each side of a flat item, without the arms touching each other.
10. A gripper lifts a thing just off what it stands on and carries it low and level along the front lane. It never carries one over the rail, over the shelf, or across the back of the bench.
11. The two arms do not hit each other. One arm working at the fold square stays out of the other arm's path to the box.
12. If the arms cannot reach a place, park the base again with the Base positioning steps until lines 8 to 11 are all true.

Materials checklist

1. The bench is one long flat top set in front of the wardrobe. Nobody moves it, leans on it, or stands anything on its edge.
2. The bench top is dry, clean, and clear but for the out stack, the open box, and the lid.
3. The fold square is the clear flat middle of the bench. It is wide enough for one item to lie out flat, with a hand's width of clear bench all around it.
4. The out stack lies flat at the front left. It holds four out-of-season items, one on top of another, each laid out flat and squared to the one under it.
5. Each of the four out items is a soft folding item: a top, a jumper, or a pair of trousers. Nothing is stiff. Nothing has a hood, a hard buckle, or an open zip.
6. Each of the four out items is clean and dry, and it is small enough to fold to a flat pack that fits in the near half of the box with a finger's width to spare on every side.
7. The storage box stands open at the front right of the bench, square to the front edge, with its lid off.
8. The box has a low divider across its middle. The half nearest the front edge is the near half. The half nearest the back is the far half.
9. The near half is empty at the start. It takes the four folded items in one stack.
10. The far half holds the four in-season items at the start, in one stack, in a fixed order. The two on top are the hang items. The two under them are the shelf items.
11. Each hang item is folded once over its own hanger, with the hook standing up clear of the fold, so a gripper closes on the hook and nothing else.

12. Each shelf item is folded flat, with no hanger in it.
13. The lid lies flat on the bench past the box, at the right end, out of the front lane. It is the right lid for this box and it sits down on all four sides.
14. The rail is a short hanging rail above the left end of the bench. It is empty at the start. It is fixed and does not turn or swing. It is low enough for a gripper to put a hook on it without stretching.
15. There is room on the rail for two hangers side by side, with a hand's width between them and a hand's width clear at each end.
16. The shelf is an open shelf above the right end of the bench. It is empty, flat, and dry at the start. It is deep enough for a folded item to lie on it with nothing hanging over the front edge.
17. The stow bay is the open bay under the shelf at the right end. It is empty at the start. It is wide enough and deep enough for the closed box to sit inside it with nothing sticking out past the front.
18. The front lane is the strip of bench along the front edge, from the out stack to the box. It is empty at the start. Nobody ever puts anything down in it.
19. The floor around the base is clear and dry.
20. Nothing else is on the bench, on the rail, on the shelf, or in the bay.

Workspace layout

Nothing is marked or taped out. You judge every place below by eye, from the bench and the wardrobe themselves.

Back row, left to right:

1. **Rail**, above the left end of the bench. Empty at the start. It holds two hangers at the end. Never moved.
2. **Shelf**, above the right end of the bench. Empty at the start. It holds two folded items at the end. Never moved.

Middle row:

1. **Fold square**, the clear middle of the bench. Every fold happens here. One item is on it at a time. Nothing is ever left on it.

Front row, left to right:

1. **Out stack**, front left. The four out-of-season items lie here in one flat stack. It empties as the box fills.
2. **Box spot**, front right. The open box stands here. Its near half takes the four folded items. Its far half holds the four in-season items at the start.
3. **Lid spot**, the right end of the bench, past the box, out of the front lane. The lid lies flat here until Step 7.

4. **Stow bay**, under the shelf at the right end. Empty at the start. The closed box goes here at the end.

Front lane: the clear strip of bench along the front edge, from the out stack to the box. Every carry goes along it. Nobody ever puts anything down in it.

No carry passes over the rail, over the shelf, or across the back of the bench, and nothing travels to the far side of the bench.

Vocabulary

Out item: one of the four out-of-season items in the out stack. All four are folded and boxed before anything comes out of the box.

In item: one of the four in-season items in the far half of the box. Two hang. Two go on the shelf.

Hang item: an in item folded over its own hanger, with the hook standing up. The two hang items are the two on top of the in stack.

Shelf item: an in item folded flat with no hanger. The two shelf items are the two under the hang items.

Hook: the curved top of a hanger. It is the only part of a hang item a gripper ever closes on.

Top of the stack: the item lying highest with nothing on it. You always take the top one. You never pull one out from under another.

Fold square: the clear middle of the bench. One item lies out flat here while it is folded. It is empty before an item comes and empty after the item leaves.

Laid out flat: the item lies on the fold square face up, spread out, with no part folded under itself, no sleeve trapped, and no bunching.

Fold: one move that brings one part of the item over onto another part and leaves it lying flat. Each item gets three folds and no more.

Flatten: the closed gripper runs once along a new fold, from one end to the other, close over the bench, to press the fold down. It does not push the item along the bench.

Square fold: the folded edges run straight, the corners line up within a finger's width, and no sleeve, hem, or corner sticks out of the pack.

Flat pack: the finished folded item. It lies flat, holds its shape when let go, and fits in the near half of the box with room on every side.

Near half: the half of the box nearest the front edge of the bench. It is empty at the start and holds the four flat packs at the end.

Far half: the half of the box nearest the back. It holds the four in items at the start and is empty at the end.

Divider: the low wall across the middle of the box that keeps the two halves apart. Nothing is ever laid across it.

Low carry: you lift the thing just off what it stands on, bring it forward into the front lane, and carry it along the lane close over the bench and level. You never carry it over the rail, over the shelf, or across the back of the bench.

Front lane: the clear strip of bench along the front edge, between the out stack and the box. Every carry goes along it. Nobody puts anything down in it.

Hanging right: the hook sits all the way over the rail, the hanger hangs straight down, the item hangs level on it, and it does not swing or slide along the rail when the gripper lets go.

Sitting right on the shelf: the flat pack lies flat on the shelf, square to the shelf's front edge, fully on the shelf with nothing hanging over any edge, and beside the other pack and not on top of it.

Lid on right: the lid lies down on the box on all four sides, square with the box, with no corner riding up and no gap you can see from the front.

Stowed right: the closed box sits flat in the stow bay, square to the bay, fully inside it, with no part sticking out past the front of the bay.

Drop: an item, a hanger, or the lid that falls on the floor. You do not pick a drop back up. You report it.

Loose item: an item that ends up on the bench outside the out stack, the fold square, the box, the rail, or the shelf. It is left where it lands and cleared in the reset, never on camera.

Back and clear: the arm is pulled back, so no part of it is over the fold square, the box, the front lane, or the bench.

Steps

Step 1: Fold the first out item and lay it in the box

Goal: the first out item lies as a flat pack in the near half of the box, and the fold square is empty.

1.1 Take the top item to the fold square

Close a gripper on the top item of the out stack, at the middle of its near edge.

Lift it straight up off the stack. It leaves the stack, and no higher.

Pull it forward into the front lane, carry it along the lane to the fold square in a low carry, and lay it down.

Check: one item lies on the fold square and the out stack is one item shorter. The item under it has not come with it and has not slid. If two items came up together, lay both back on the stack, square them, and take the top one again.

1.2 Lay it out flat

Open the item out on the fold square, face up, so it lies spread out and flat.

Pull the sleeves and the corners out, so no part is folded under itself.

Square it to the front edge of the bench.

Check: the item lies flat, face up, and square, with nothing folded under it and no sleeve trapped. The whole item is on the fold square. If a part is under itself, lift that part, pull it clear, and lay it flat again.

1.3 Fold it three times

Fold the left side in to the middle. Flatten the new fold.

Fold the right side in to the middle. Flatten the new fold.

Fold the bottom half up onto the top half. Flatten the new fold.

Check: the pack is a square fold. Its edges run straight, its corners line up, and no sleeve or corner sticks out. It is three folds, not two and not four. If a corner sticks out, open that last fold, line the edge up, and fold it again. Make no more than two extra tries on one item.

1.4 Lay it in the near half

Lift the flat pack straight up off the fold square, with a gripper on each side of it, so it keeps its shape.

Carry it along the front lane to the box in a low carry. Keep it level the whole way.

Lower it straight down into the near half of the box, flat and square, and open the grippers.

Check: the pack lies flat in the near half. It is fully inside that half, it does not lie across the divider, and it has not fallen open. The far half has not been touched. If it lies across the divider, lift it straight up, move it over, and lower it again.

Expected state: one flat pack lies in the near half of the box. The out stack holds three items. The fold square is empty. The far half still holds all four in items. Both grippers are back and clear.

Step 2: Fold the other three out items the same way

Goal: all four out items lie as flat packs in the near half of the box, and the out stack is empty.

2.1 Do the same for the second item

Do 1.1, 1.2, 1.3, and 1.4 again, for the item now on top of the out stack.

Lay the pack flat in the near half, on top of the first pack, square with it.

Check: two packs lie in the near half, one on the other, square, and both fully inside that half. The out stack holds two items.

2.2 Do the same for the third item

Do 1.1, 1.2, 1.3, and 1.4 again, for the item now on top of the out stack.

Lay the pack flat in the near half, on top of the second pack, square with it.

Check: three packs lie in the near half, one on another, square, and all fully inside that half. The out stack holds one item.

2.3 Do the same for the fourth item

Do 1.1, 1.2, 1.3, and 1.4 again, for the last item on the out stack.

Lay the pack flat in the near half, on top of the third pack, square with it.

Check: four packs lie in the near half, one on another, square, and all fully inside that half. The out stack spot is empty bench. The stack of four sits below the top of the box, so the lid will go on later. If the stack stands proud of the box, press the top pack down once, flat and gentle. Do not push the box.

Expected state: four flat packs lie in the near half of the box. The out stack spot is empty. The fold square is empty. The four in items still lie in the far half. Both grippers are back and clear.

Step 3: Hang the first in item

Goal: the first hang item hangs on the rail, straight and level, and the far half holds three items.

3.1 Lift it out by the hook

Close a gripper on the hook of the top item in the far half. Close on the hook and nothing else.

Lift it straight up out of the box, until the item hangs clear of the box and the divider.

Do not let the item drag on the box edge. Do not let it swing.

Check: the item hangs from the hook, straight and clear of the box. Nothing came up with it. The packs in the near half have not moved. If a second item came up with it, lower both back into the far half, let go, and take the top hook again.

3.2 Carry it to the rail

Carry it along the front lane to the left end of the bench in a low carry. Keep the hook level and the item hanging straight.

Keep the item clear of the bench top, the box, and the lid. Do not let it brush the shelf.

Bring it in front of the rail. Then lift the hook up to the rail.

3.3 Put the hook on the rail

Lower the hook onto the rail, all the way over it, near the left end of the rail.

Let the hanger settle and hang straight down.

Open the gripper and pull it back away from the hanger.

Check: the item is hanging right. The hook is all the way over the rail, the hanger hangs straight, the item hangs level, and it does not swing or slide when the gripper lets go. If the hook sits on the edge of the rail, or the item hangs crooked, close on the hook again, lift it just off the rail, and put it on again.

Expected state: one item hangs on the rail. The far half holds three items. The near half holds four packs. Both grippers are back and clear.

Step 4: Hang the second in item

Goal: the second hang item hangs on the rail beside the first, and the far half holds two items.

4.1 Lift it out by the hook

Close a gripper on the hook of the item now on top of the far half. Close on the hook and nothing else.

Lift it straight up out of the box, until the item hangs clear of the box and the divider.

Do not let the item drag on the box edge. Do not let it swing.

Check: the item hangs from the hook, straight and clear of the box. Nothing came up with it. If it catches on the box edge, lower it back in, let go, take the hook again, and lift straighter.

4.2 Carry it to the rail

Carry it along the front lane to the left end of the bench in a low carry. Keep the hook level and the item hanging straight.

Keep it clear of the item already on the rail. Do not knock it or push it along the rail.

Bring it in front of the rail. Then lift the hook up to the rail.

4.3 Put the hook on the rail

Lower the hook onto the rail, all the way over it, a hand's width to the right of the first hanger.

Let the hanger settle and hang straight down.

Open the gripper and pull it back away from the hanger.

Check: the item is hanging right, and the two items hang apart and do not touch. The first item has not moved along the rail. If the two touch, close on the second hook, lift it just off the rail, and put it on again further along.

Expected state: two items hang on the rail, apart and straight. The far half holds two items, both folded flat with no hanger. The near half holds four packs. Both grippers are back and clear.

Step 5: Put the first shelf item on the shelf

Goal: the first shelf item lies flat on the shelf, square and fully on it, and the far half holds one item.

5.1 Lift it out flat

Close a gripper on each side of the top item in the far half, at the middle of its sides.

Lift it straight up out of the box, keeping it flat and level, so it does not fall open.

Do not let it drag on the box edge or on the divider.

Check: the pack comes out flat and holds its shape. Nothing came up with it. If it falls open in the air, lower it back into the far half, let go, close on it again lower down, and lift it again.

5.2 Carry it to the shelf

Carry it along the front lane to the right end of the bench in a low carry. Keep it flat and level the whole way.

Do not carry it over the box or over the lid.

Bring it in front of the shelf. Then move it straight back, over the shelf.

5.3 Lay it on the shelf

Lower it straight down onto the shelf, to the left side of the shelf.

Lay it flat, square to the shelf's front edge, fully on the shelf.

Open the grippers and pull them back away from the item.

Check: the item is sitting right on the shelf. It lies flat, square, and fully on, with nothing hanging over any edge, and it does not slide when the grippers let go. If a corner hangs over the front edge, close on it again, lift it just off the shelf, and lay it down again further back.

Expected state: one folded item lies on the shelf. Two items hang on the rail. The far half holds one item. The near half holds four packs. Both grippers are back and clear.

Step 6: Put the second shelf item on the shelf

Goal: the second shelf item lies flat beside the first, the far half is empty, and the box is ready to close.

6.1 Lift it out flat

Close a gripper on each side of the last item in the far half, at the middle of its sides.

Lift it straight up out of the box, keeping it flat and level, so it does not fall open.

Do not let it drag on the box edge or on the divider.

Check: the pack comes out flat and holds its shape. The far half is now empty, with nothing left in the bottom of it. If something is left behind, lift it out the same way before you go on.

6.2 Carry it to the shelf

Carry it along the front lane to the right end of the bench in a low carry. Keep it flat and level the whole way.

Do not carry it over the box or over the lid.

Bring it in front of the shelf. Then move it straight back, over the shelf.

6.3 Lay it beside the first

Lower it straight down onto the shelf, to the right of the first item, a hand's width away from it.

Lay it flat, square to the shelf's front edge, fully on the shelf.

Open the grippers and pull them back away from the item.

Check: the item is sitting right on the shelf, beside the first item and not on top of it. The first item has not moved. If the two touch or one sits on the other, close on the second one, lift it just off the shelf, and lay it down again beside the first.

Expected state: two folded items lie side by side on the shelf. Two items hang on the rail. The far half of the box is empty. The near half holds four packs. Both grippers are back and clear.

Step 7: Close the box

Goal: the lid is on the box, square and down on all four sides.

7.1 Look in the box

Look into the box from the front.

Count four flat packs in the near half. Count nothing in the far half.

Check: the near half holds four packs and the far half is empty. If the far half is not empty, do Step 5 or Step 6 again for what is left before you close the box. Do not put the lid on a box with an item still in it.

7.2 Pick up the lid

Close a gripper on each side of the lid, at the middle of its sides.

Lift the lid straight up off the bench. It leaves the bench, and no higher.

Carry it along the front lane to the box in a low carry. Keep it flat and level.

Check: the lid is held flat and level. Nothing came up under it. The box has not moved.

7.3 Put the lid on

Bring the lid over the box, lined up with it on all four sides.

Lower it straight down onto the box until it sits down on all four sides.

Open the grippers and pull them back away from the lid.

Check: the lid is on right. It is square with the box, down on all four sides, with no corner riding up and no gap you can see from the front. If a corner rides up, close on the lid again, lift it just off, line it up, and lower it again.

Expected state: the box stands closed at the front right of the bench. The bench top is clear but for the box. Both grippers are back and clear.

Step 8: Stow the box and end the episode

Goal: the box sits square in the stow bay, and the wardrobe is left as the episode wants it.

8.1 Lift the closed box

Close a gripper on each side of the box, low down on the box body, below the lid.

Do not hold it by the lid. Do not hook a gripper under the lid edge.

Lift the box straight up off the bench. It leaves the bench, and no higher.

Check: the box is held level, by its body and not by its lid. The lid has not lifted off. If the lid lifts, lower the box, put the lid back on, take the body again lower down, and lift again.

8.2 Carry it to the bay

Carry the box along the front lane to the right end of the bench in a low carry. Keep it level the whole way.

Do not carry it over the shelf. Do not knock the items on the shelf.

Bring it in front of the stow bay.

8.3 Put it down in the bay

Lower the box straight down into the stow bay.

Put it down flat, square to the bay, fully inside it.

Open the grippers and pull them back away from the box.

Check: the box is stowed right. It sits flat, square, fully inside the bay, with no part sticking out past the front. If it sits crooked or sticks out, close on the box body again, lift it just off the bay floor, and put it down again.

8.4 End the episode

Check the rail: two items hang on it, straight, level, apart, and not swinging.

Check the shelf: two folded items lie flat on it, side by side, square, fully on the shelf.

Check the box: it is closed, and it sits square and fully inside the stow bay.

Check the bench: the out stack spot, the fold square, the front lane, the box spot, and the lid spot are all empty and clear.

Check the fixtures: the bench, the rail, the shelf, and the bay stand where they started.

Check the base has not moved.

Fix any failed check before you end.

Send both arms home with the grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the two items off the rail. Leave the rail empty.

2. Take the two folded items off the shelf. Leave the shelf flat, dry, and empty.
3. Lift the box out of the stow bay and stand it back at the box spot at the front right of the bench, square to the front edge.
4. Take the lid off and lay it flat past the box, at the right end, out of the front lane.
5. Take the four flat packs out of the near half. Leave the near half empty.
6. Shake the four packs out flat and lay them at the front left as the new out stack, one on another, each squared to the one under it.
7. Fold each of the two rail items once over its own hanger, hook standing up clear of the fold, and lay both in the far half of the box.
8. Fold each of the two shelf items flat and lay both in the far half, under the two on hangers, so the two on hangers are on top.
9. Check the far half now holds four items in the fixed order: two on hangers on top, two folded flat under them.
10. Wipe the bench top and dry it. Leave it clear but for the out stack, the open box, and the lid.
11. Change the box if a corner is crushed, if the divider is split or loose, or if the lid will not sit down on all four sides.
12. Change a hanger if its hook is bent, if it is too wide for the rail, or if an item slides off it.
13. Change an item if it is torn, damp, or so creased that it will not fold to a flat pack.
14. Check the rail is fixed and does not turn or swing, and the shelf is level and does not tilt.
15. Check the stow bay is empty and clear, and wide enough and deep enough for the closed box.
16. Pick up anything that fell on the floor and check the floor is clear and dry.
17. Check the base is still locked and parked square. Then do the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP. They are reviewed side by side in the review tool.

How to record a violation in review

For each violation, write down the start timestamp, the violation name, and the SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining. It is not an annotation label.

Episode handling

Tag every violation with its timestamp and its name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

Visible cue: the bench moves in the frame, the bench edge changes angle in the frame, or the base rolls, slides, or turns at any time after recording starts.

SOP rule broken: Steps 1 to 8. The base is parked and locked before recording and stays still for the whole episode.

Coaching note: park it, lock it, push it to test it, then start recording. A base that moves is a lost episode.

Violation: Lifted high or carried over a fixture

Visible cue: an item, the lid, or the box is lifted well above the bench instead of just off what it stands on, or is carried over the rail, over the shelf, or across the back of the bench. The one lift that may go up is the hook going onto the rail.

SOP rule broken: Steps 1.1, 1.4, 3.2, 4.2, 5.2, 6.2, 7.2, 8.2. Lift just off, come forward into the front lane, and carry low and level.

Coaching note: up a little, forward, then along. Only the hook goes up, and only at the rail.

Violation: Carried outside the front lane

Visible cue: a carry goes across the back of the bench instead of along the front lane. Or a thing is put down in the lane in the middle of a carry.

SOP rule broken: Steps 1.1, 1.4, 3.2, 4.2, 5.2, 6.2, 7.2, 8.2. Every carry goes along the front lane, and nothing is ever put down in it.

Coaching note: the lane is the road. Nothing crosses the back of the bench, and nothing parks in the lane.

Violation: Leaned on or moved a fixture

Visible cue: a gripper, wrist, or forearm rests on the bench edge, the rail, the shelf, or the bay. Or the rail turns or swings. Or the shelf tilts. Or the box slides on the bench under a set down.

SOP rule broken: Steps 1 to 8. Nobody leans on, pushes, drags, or lifts the bench, the rail, the shelf, or the bay.

Coaching note: the arm holds itself up. If something moves when you touch it, you touched it too hard.

Violation: Dragged instead of lifted

Visible cue: an item, the lid, or the box is slid or dragged along the bench, along the shelf, or along the rail instead of being lifted straight up and carried.

SOP rule broken: Steps 1 to 8. Every pick lifts straight up off the surface before it moves anywhere.

Coaching note: lift first, then move. A drag creases the item and pushes the fixture.

Violation: Held by the wrong part

Visible cue: a gripper closes on a sleeve, a hem, or a corner to lift a whole item. Or it closes on the garment of a hang item instead of on the hook. Or it holds the closed box by the lid or hooks under the lid edge.

SOP rule broken: Steps 1.1, 3.1, 4.1, 5.1, 6.1, 8.1. Take a flat item at the middle of its sides, a hang item by its hook, and the box by its body below the lid.

Coaching note: hook for hangers, sides for packs, body for the box. Never the lid, never a sleeve.

Violation: Took the wrong item off a stack

Visible cue: a gripper pulls an item out from under another one, or lifts two items at once, on the out stack or in the far half of the box.

SOP rule broken: Steps 1.1, 2.1, 2.2, 2.3, 3.1, 4.1, 5.1, 6.1. You always take the top one.

Coaching note: top one only. If two come up, put both back and start that pick again.

Violation: Folded somewhere other than the fold square

Visible cue: an item is folded in the air, on the out stack, on the shelf, or inside the box, instead of lying out flat on the fold square.

SOP rule broken: Steps 1.2 and 1.3. The item lies out flat on the fold square before any fold is made.

Coaching note: lay it out first, then fold it. Nothing is ever folded hanging in the air.

Violation: Not laid out flat before folding

Visible cue: the first fold is made while a sleeve is trapped, a corner is folded under, or the item is still bunched from the stack.

SOP rule broken: Step 1.2. Pull the sleeves and the corners out and lay the item flat and square before folding.

Coaching note: flat and square first. A fold over a trapped sleeve never comes out square.

Violation: Wrong number of folds

Visible cue: an item goes into the box after two folds, or after four or more folds.

SOP rule broken: Step 1.3. Each item gets three folds: left in, right in, bottom up.

Coaching note: three folds, every item, every time. Count them as you make them.

Violation: Fold not square

Visible cue: the flat pack has edges that run crooked, corners that miss each other by more than a finger's width, or a sleeve, hem, or corner sticking out of the pack.

SOP rule broken: Step 1.3. The pack is a square fold before it leaves the fold square.

Coaching note: line the edge up before you bring it over. Fix it on the square, not in the box.

Violation: Pack put in the wrong half

Visible cue: a folded out item is laid in the far half of the box, or laid across the divider, or laid on top of the in items.

SOP rule broken: Step 1.4. The four flat packs go in the near half and nowhere else.

Coaching note: near half is out, far half is in. Nothing lies across the divider.

Violation: Out items and in items mixed

Visible cue: an in item is left in the near half, or a flat pack ends up in the far half, or the two are stacked together at any time.

SOP rule broken: Steps 1.4, 3.1, 4.1, 5.1, 6.1. The two halves are never mixed.

Coaching note: two halves, two jobs. What comes out never goes back in.

Violation: Done in the wrong order

Visible cue: an in item is taken out of the box before all four out items are folded and in the near half. Or a shelf item is taken out before both hang items are on the rail. Or the lid goes on before the far half is empty.

SOP rule broken: Steps 1 to 7. Four out first, then the two hang items, then the two shelf items, then the lid.

Coaching note: fill the box, empty the box, then close it. One thing at a time, in order.

Violation: Item not hanging right

Visible cue: the hook sits on the edge of the rail instead of all the way over it, the hanger hangs crooked, the item hangs off one shoulder, or the hanger swings or slides along the rail when the gripper lets go.

SOP rule broken: Steps 3.3 and 4.3. The hook goes all the way over the rail and the hanger settles straight before the gripper lets go.

Coaching note: all the way over, then watch it settle before you let go.

Violation: Hangers touching or knocked

Visible cue: the two hangers end up touching, or the second hanger pushes the first one along the rail, or the first item is knocked off the rail while the second goes on.

SOP rule broken: Step 4.3. The second hook goes on a hand's width from the first, without touching it.

Coaching note: leave a gap. The rail has room for two, so use it.

Violation: Item not sitting right on the shelf

Visible cue: a folded item hangs over the front edge or a side edge of the shelf, lies crooked, sits on top of the other item, or slides when the grippers let go.

SOP rule broken: Steps 5.3 and 6.3. Lay it flat, square, and fully on the shelf, beside the other item.

Coaching note: fully on and square, side by side. Nothing hangs over an edge.

Violation: Pack fell open

Visible cue: a flat pack comes apart in the air, in the box, or on the shelf, so it no longer holds its folded shape.

SOP rule broken: Steps 1.4, 5.1, 5.3, 6.1, 6.3. Lift a pack with a gripper on each side and keep it flat and level.

Coaching note: two sides, level, no swing. A pack lifted from one corner always opens.

Violation: Lid put on wrong

Visible cue: the lid ends up crooked on the box, riding up on a corner, with a gap you can see from the front, or resting half on the box and half on the bench.

SOP rule broken: Step 7.3. The lid goes down on all four sides and square with the box.

Coaching note: line it up over the box before you lower it. Straight down, all four sides.

Violation: Box closed too early

Visible cue: the lid goes on while an in item is still in the far half, or while a flat pack is still on the bench or on the fold square.

SOP rule broken: Step 7.1. Look in the box and count four packs in the near half and nothing in the far half before the lid goes on.

Coaching note: look in first. A lid on a wrong box hides the mistake.

Violation: Box stowed wrong

Visible cue: the box ends up crooked in the bay, sticking out past the front of the bay, on its side, on the bench, or on the floor.

SOP rule broken: Step 8.3. The box sits flat, square, and fully inside the stow bay.

Coaching note: straight down into the bay, square, and all the way in.

Violation: Something dropped or knocked over

Visible cue: an item, a hanger, the lid, or the box slips out of a gripper, falls on the bench or on the floor, or is knocked off the rail or off the shelf by an arm going past.

SOP rule broken: Steps 1 to 8. One thing moves at a time, and each one is put down under control.

Coaching note: close all the way before you lift, and put it down before you let go.

Violation: Item touched the floor

Visible cue: any part of an item, a hanger, the lid, or the box touches the floor at any time.

SOP rule broken: Steps 1 to 8. Clothes go on the bench, in the box, on the rail, or on the shelf, and nowhere else.

Coaching note: stop, report it, and reset. You do not pick a drop back up in the middle of an episode.

Violation: Failed check not retried

Visible cue: a check in a step clearly fails, and the episode carries on with no retry. For example, a sleeve sticks out of a pack, a pack lies across the divider, a hook sits on the edge of the rail, or a folded item hangs over the shelf edge.

SOP rule broken: Steps 1 to 8. Do the retry written under each failed check before you move on.

Coaching note: a check is only worth doing if you do the retry. Fix it where it happened.

Violation: Wrong ending

Visible cue: the episode ends with an item out of place, a pack still on the bench, the box open, the box out of the bay, or the fold square or the front lane not clear. Or an arm is not home, or a gripper is not fully open.

SOP rule broken: Step 8.4. Check the rail, the shelf, the box, the bench, the fixtures, and the base. Then send both arms home with the grippers open and stop recording.

Coaching note: check first. Going home is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, throw the episode away, and never use them for coaching.

Recording stopped or paused during the episode (recording system).

Camera dropped frames or lost its feed (capture system).

Camera cannot see inside the box, cannot see the rail, or cannot see the front edge of the shelf, so nobody can tell if an item was placed right.

Hardware fault on an arm: gripper failure, drift, a crash caused by the controller, or a motor error.

Base fault: a lock that will not hold, or a base that rolls, slides, or turns with the lock on.

Faulty box: a crushed corner, a split or loose divider, a bottom that folds open, or a box that will not stand square on the bench.

Faulty lid: one that will not sit down on all four sides on a correctly placed box, or one that is the wrong size for the box.

Faulty hanger: a bent hook that will not sit over the rail, a hanger too wide for the rail, or one that lets the item slide off in a correct hang.

Faulty rail: one that turns, swings, sags, or comes loose under a correctly placed hook.

Faulty shelf: one that tilts, that is too shallow for a folded item to lie fully on, or that is wet or dirty at the start.

Faulty stow bay: one too small for the closed box, or one blocked by something that was there before the episode started.

Faulty item: one that is torn, damp, stiff, or so creased that it will not fold to a flat pack, or one too big to fit the near half of the box.

The far half arrives in the wrong order, so the two on hangers are not the two on top.

A place turns out to be too far for the arm with the base parked correctly, so the out stack, the fold square, either half of the box, the lid, the rail, the shelf, or the bay cannot be reached without stretching or folding the arm.

Annotation subtasks (from SOP)

1. Take the top out item to the fold square
2. Lay the out item flat on the fold square
3. Fold the out item three times
4. Lay the flat pack in the near half of the box
5. Repeat the fold and box for the second out item
6. Repeat the fold and box for the third out item
7. Repeat the fold and box for the fourth out item
8. Lift the first hang item out of the box by its hook
9. Carry the first hang item along the front lane to the rail
10. Put the first hook on the rail
11. Lift the second hang item out of the box by its hook
12. Carry the second hang item along the front lane to the rail
13. Put the second hook on the rail
14. Lift the first shelf item out of the box flat
15. Lay the first shelf item on the shelf
16. Lift the second shelf item out of the box flat
17. Lay the second shelf item beside the first
18. Pick up the lid and put it on the box
19. Lift the closed box and stow it in the bay
20. Send both arms home and end the episode